PhonePe Transaction Data Analysis Dashboard

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Tools Used: Python, MySQL, Pandas, Plotly, Streamlit, GitHub

1. Project Overview

This project analyzes digital payment data from the PhonePe ecosystem to understand transaction trends, user growth, device dominance, insurance adoption, and market expansion across India. The insights are visualized using an interactive Streamlit dashboard.

2. Project Objectives

- Analyze digital payment transaction growth.
- Identify top performing states and districts.
- Understand mobile device dominance.
- Study digital insurance growth trends.
- Explore regional market expansion and user growth.

3. Dataset Description

aggregated_transactions: Transaction count and amount by state, year, and quarter.

aggregated_user: Mobile brand usage and user counts.

aggregated_insurance: Insurance transaction amounts and counts.

map_transaction: District level transaction data.

top_user: Registered users by pincode.

4. SQL Analysis Queries

Transaction Dynamics

SELECT year, SUM(transaction_amount) FROM aggregated_transactions GROUP BY year ORDER BY year;

SELECT quarter, SUM(transaction_amount) FROM aggregated_transactions GROUP BY quarter ORDER BY quarter;

SELECT state, SUM(transaction_amount) FROM aggregated_transactions GROUP BY state ORDER BY SUM(transaction_amount) DESC LIMIT 10;

SELECT transaction_type, SUM(transaction_count) FROM aggregated_transactions GROUP BY transaction_type;

SELECT year, SUM(transaction_amount) FROM aggregated_transactions GROUP BY year ORDER BY SUM(transaction_amount) DESC LIMIT 1;

Device Dominance

SELECT brand, SUM(user_count) FROM aggregated_user GROUP BY brand ORDER BY SUM(user_count) DESC;

SELECT brand, SUM(user_count) FROM aggregated_user GROUP BY brand ORDER BY SUM(user_count) DESC LIMIT 5;

SELECT year, brand, SUM(user_count) FROM aggregated_user GROUP BY year, brand;

SELECT quarter, SUM(user_count) FROM aggregated_user GROUP BY quarter;

SELECT brand, SUM(user_count) FROM aggregated_user GROUP BY brand ORDER BY SUM(user_count) DESC LIMIT 1;

Insurance Growth

SELECT year, SUM(insurance_amount) FROM aggregated_insurance GROUP BY year ORDER BY year;

SELECT quarter, SUM(insurance_amount) FROM aggregated_insurance GROUP BY quarter;

SELECT state, SUM(insurance_amount) FROM aggregated_insurance GROUP BY state ORDER BY SUM(insurance_amount) DESC LIMIT 10;

SELECT SUM(insurance_count) FROM aggregated_insurance;

SELECT year, SUM(insurance_amount) FROM aggregated_insurance GROUP BY year ORDER BY SUM(insurance_amount) DESC LIMIT 1;

Market Expansion

SELECT district, SUM(transaction_amount) FROM map_transaction GROUP BY district ORDER BY SUM(transaction_amount) DESC LIMIT 10;

SELECT district, SUM(transaction_count) FROM map_transaction GROUP BY district;

SELECT year, SUM(transaction_amount) FROM map_transaction GROUP BY year ORDER BY year;

SELECT district, SUM(transaction_amount) FROM map_transaction GROUP BY district ORDER BY SUM(transaction_amount) DESC LIMIT 1;

SELECT COUNT(DISTINCT district) FROM map_transaction;

User Growth

SELECT SUM(registered_users) FROM top_user;

SELECT pincode, SUM(registered_users) FROM top_user GROUP BY pincode ORDER BY SUM(registered_users) DESC LIMIT 10;

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SELECT year, SUM(registered_users) FROM top_user GROUP BY year ORDER BY year;

SELECT quarter, SUM(registered_users) FROM top_user GROUP BY quarter;

SELECT pincode, SUM(registered_users) FROM top_user GROUP BY pincode ORDER BY
SUM(registered_users) DESC LIMIT 1;
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5. Key Insights

- Digital transactions increased significantly over the years.
- Certain states contribute a major share of total transactions.
- Mobile usage is dominated by a few major smartphone brands.
- Insurance adoption is steadily growing.
- Urban districts generate higher transaction volumes.

6. Recommendations

- Expand fintech awareness in low adoption regions.
- Promote digital insurance services through incentives.
- Partner with smartphone brands to increase user adoption.
- Improve rural fintech infrastructure.

7. Conclusion

The analysis highlights the rapid growth of digital payment adoption in India. The dashboard enables interactive exploration of transaction trends, user behavior, and market expansion across regions.